Conceptual Design for a BSL-3 Suitable Stage-Top Bio-containment chamber

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Chapter 1

Introduction

1.1 Background and Summary

Long term Live-Cell imaging is a very useful research tool when determining long term behavior of cell cultures in various situations. Commercial solutions are unable to provide adequate containment for the use within a Bio-Safety Level 3 (BSL-3) and often require the use of a separate containment enclosure, which introduces additional costs and complexity.

This project focuses on the conceptual design of a low-cost Stage-Top Incubator that is capable of being utilized within a BSL-3 laboratory, without the requirement for additional containment. The primary focus with this project is specific suitability with Mycobacterium tuberculosis, however does consider other microbe requirements

1.2 Scope and Limitations

This project is limited to the following:

- Preliminary Research
- Temperature Control
- Compatibility with Multi-Well microtiter plates
- Primary compatibility with Mycobacterium tuberculosis (M. tb).

This project is limited to compatibility with a BSL-3 Lab and does not consider higher containment requirements. The environmental control aspects of the incubator are limited to just temperature control for the preliminary conceptual design. The initial design is limited to compatibility with a multi-well microtiter plate. The use case of this project is limited to that of Mycobacterium tuberculosis.

Chapter 2

Research

The conceptual design began with assessing existing commercial products to identify the key environmental control specifications required for the incubator to function correctly. Additionally, this was useful in identifying why the commercial products are incompatible with a BSL-3 lab when used without a separate external biological containment mechanism.

2.1 Zeis/Pecon

The Zeis/Pecon stage top incubator utilities four modules to control the environmental conditions within the incubator stage.

The Pecon Temperature Module S[1] regulates the temperature of the incubator. It has four 8-pin connectors that are used for temperature control of heated components such as the Incubator lid, incubator base, microscope objective and humidifier device. The temperature module is the base module for the entire incubation system, providing electrical power and communications to all connected devices. It directly connects to the CO2 module to provide power, communication and makes a connection to the gas circulation system to heat the gas flow provided to the incubation chamber. The temperature module requires an external device such as a PC or compatible microscope for operation as it does not have any control panel. The module maintains culture temperature by monitoring and controlling the temperature of the incubator base, lid and optionally the microscope objective. The module infers the culture temperature using pre-set temperatures, or by performing a pre-experiment calibration whereby an identical set-up is placed in the chamber with the included culture temperature sensor to tune the required chamber temperatures.

The Pecon CO2 Module S[2] attaches to the temperature module for electrical power and communication, and provides CO2 concentration control to the airflow passed into the incubation chamber. The CO2 module circulates gas throughout all the connected units and mixes the gas before pumping the resultant mixture into the incubation chamber. The module contains a CO2 sensor which monitors the concentration of CO2 in the gas mixture, and controls the release of pure CO2 into the mixing chamber in order to maintain the desired CO2 concentration. Due to the method of operation, it requires a >95% purity CO2 supply.

The Pecon O2 Module S[3] attaches to the CO2 module for electrical power and communication. This module functions similarly to the CO2 module, however it regulates the concentration of O2 present in the gas mixture by controlling the release of a pure N2 supply into the air mixture to displace the O2 present. This method of O2 displacements means that the O2 module is only capable of providing hypoxia conditions by lowering O2 concentration. Due to the method of operation, it requires a >99% purity N2 supply.

Both the CO2 module and O2 module are supplied with a humidifying bottle which functions by bubbling the the gas airflow thorugh distilled water to raise the relative humidity (RH). The Pecon Heating Device Humidity S1[4] is an optional module which connects to the Temperature module to heat the humidifying bottle to increase the RH and maintain adequete gas temperature provided to the incubation chamber

The incubator stage includes a heated glass topped lid[5] and a heated insert which accepts mounts for the different culture vessels used in microscopy. The heated insert directly heats the culture medium through physical contact, and the heated lid provides radiative heating and condensation prevention by heating the glass panel directly. The insert supports perfusion experiments utilising two ports which allow tubes to be run to the culture.

2.2 Ibidi

The Ibidi stage top incubator functions similarly to the Zeis/Pecon module, however it makes use of three modules to maintain environmental conditions.

The Heating System Silver Line[6] module regulates the temperature of the incubator. It makes use of 6 independent channels connected via 4 ports, two of which connect to two channels each. These channels are used to monitor and control the temperature of the components used in the system, such as the heated lid and heated plate. The module maintains culture temperature by monitoring and controlling the temperature of the heated lid, heated base and optionally the microscope objective. The module infers the culture temperature using pre-set temperatures, or by performing a calibration whereby an identical set-up is placed in the chamber with the included thermocouple to tune the required chamber settings to meet the desired culture temperature. The temperature module can be operated using the on device buttons and screen, or it can be connected to a PC for control using the free software. The temperature module does not heat the gas mixture.

The Gas Incubation System[7] comes in two different models, the CO2 - Silver Line and the CO2/O2 - Silver Line. The difference between these two systems is the former is only capable of regulating CO2 concentration, where as the latter has the ability to regulate O2 concentration as well as CO2 concentration. The module has three gas inputs for pressurised air, CO2 and N2. The CO2 and N2 are fed into and mixed with the pressurised air to obtain a mixture with the desired CO2 and O2 concentrations. The module then heats the gas mixture before pumping it to the incubation chamber. If the CO2 only module is used, the N2 air input is still present on the device, however it is sealed internally and so the module is unable to regulate O2 concentration. Due to the method of using N2 to displace O2 in the mixture, this system is only capable of producing hypoxia environments. The gas incubator module has two output ports which are used in conjunction with the Humidifying column to mix the dry gas with humidified gas to obtain the desired relative humidity as shown in Figure 2.1. The gas incubation module has two electrical connections, one to control the heating of the humidifying column and gas line, and the second to connect to the humidity sensor present in the incubation chamber. The module can be operated using the on device buttons and screen, or it can be connected to a PC for control using the free software.

The Humidifying column[7] works by bubbling the airflow through heated, distilled water to increase

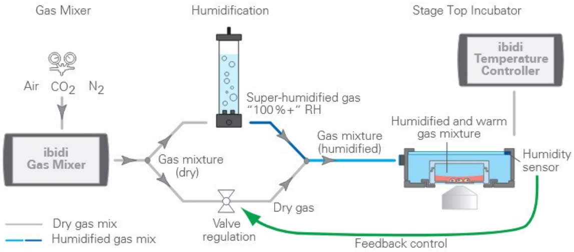


Figure 2.1: Diagram Showing the Humidity Control Mechanism Used by the Ibidi Gas Incubation System [7]

the relative humidity to approximately 100% RH. This is then mixed with dry gas to obtain the desired % RH.

The Ibidi incubation chamber comes in two versions, one version supports culture mediums with Petri dishes and slides, the second version supports multi-well plates. The Petri dish/Slide version of the chamber supports perfusion experiments using two ports which allow tubes to be run to the culture. Both chambers directly heat the culture medium through physical contact and provide radiative heating and condensation prevention by heating the glass lid.

2.3 Tokai-Hit

The Tokai-Hit system functions similarly to the Pecon and Ibidi systems however it utilises a single module to regulate the environmental conditions for temperature and CO2. Humidity is controlled by a heated water bath inside the chamber, and O2 control is achieved by utilising an optional module which then overrides all gas control.

The Tokai-Hit STXG/STXF[8] temperature and CO2 controller regulates both temperature and CO2 concentration in the incubation chamber. The controller utilises a single connector to control all component temperatures such as the Lid heater, bath heater, stage heater and optional microscope objective heater. The module infers the culture temperature using pre-set temperatures, however it can operate in feedback mode whereby a sensor directly monitors the temperature of the culture medium and automatically adjusts the component temperatures to meet the desired set point. The module can be operated using the built in touch screen or it can be connected to a PC for control using software.

The STXG and STXF models both provide airflow to the incubation chamber, however the STXG model is capable of controlling the CO2 concentration by feeding supplied CO2 into the gas mixture until the desired concentration is reached, then pumps the mixture to the incubation chamber. The STXF model is unable to regulate CO2 concentration, and so requires a pre-mixed pressurised CO2/air.

The incubator chamber uses a single design that is compatible with most major culture vessels including Petri dishes, slides and multi-well plates. This chamber supports perfusion experiments, drug administration and direct culture temperature measurement via access ports designed into the lid. The humidity inside the incubation chamber is maintained via a water bath that is present inside the chamber, and which is heated by the STXG/STXF module.

The STXG/STXF modules can optionally be connected to a separate CO2/O2 unit which overrides the gas control mixture and allows for the optional control of O2 concentration.

2.4 OkoLab

The OkoLab utilises three modules to regulate environmental conditions within the incubation chamber

The OkoLab H301-T-UNIT-BL-PLUS[9] unit is responsible for maintaining the thermal conditions within the chamber enclosure. It operates by monitoring and controlling the temperature of the chamber base, lid and optionally the microscope objective. This in turn heats the sample to the desired temperature. This heating unit is capable of sample temperature feedback utilising a thin thermocouple that can be placed in distilled water near the sample to obtain more accurate sample temperature control. Otherwise the unit is capable of using pre-sets/calibrations to infer sample temperature based on enclosure temperatures. The heating unit can be operated via a computer using available software or using optional touch screen accessory. The heating unit is capable of interfacing with the Oko-Lab Bold Line Gas Controllers. The temperature module supports a programmed temperature cycle mode where it cycles between set temperatures.

The CO2-O2-UNIT-BL[10] modules come in a variety of formats, differing in the O2 control ranges they support. The CO2-O2-UNIT-BL [0-20; 1-95] variant supports O2 concentration range from 1% to 95%. This model requires a pressurised supply of CO2, N2 and O2 to obtain the desired gas mixture, alternatively compressed air or an air pump can be connected to the N2 supply which limits the functionality to just CO2 concentration control. The CO2-O2 BL line all have the capability to directly interface with the Temperature unit to allow communication between the two units.

The HM-ACTIVE[11] operates by passing the gas mixture over heated water to increase the RH. The water temperature can be altered to control the RH of the out flowing gas. The humidity control unit is controlled by the gas mixture unit.

The H301-K-FRAME[12] uses a design that is compatible with most major culture vessels, each using a different sample holder. The chamber is also designed to work with various lids depending on the required application. The chamber supports perfusion experiments and microfluidic experiments. The chamber directly heats the culture medium through physical contact and provides radiative heating and condensation prevention by heating the glass lid

2.5 Shortfalls

2.5.1 Containment

The commercial solutions provided by Zeis/Pecon, ibidi, Tokai Hit and OkoLab are all unable to operate independently, without an external containment chamber, within a BSL-3 laboratory due to a fundamental design choice present in all solutions. The Incubation chambers are designed to have an imperfect containment seal to allow the incoming gas flows to passively exhaust out the base and any gaps in the enclosure. This can be seen in the diagram shown in Figure 2.2 which is provided by ibidi.

This means that the exhausting air is unfiltered and may be contaminated with the cells present during the experiment, which can lead to contamination of the surrounding workspace. For this reason, when

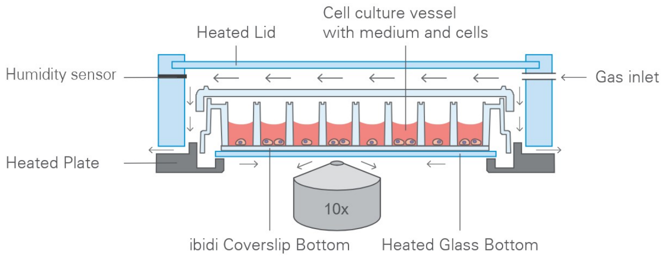


Figure 2.2: Diagram Showing the Gas Flow within the ibidi Multi-Well Incubation Chamber

working with potentially lethal pathogens, it is required that the incubator be placed within a larger containment vessel to ensure that contamination does not occur.

2.5.2 Disinfection and sterilisation

All equipment inside of a BSL-3 lab requires either sterilisation or disinfection depending on the equipment use or properties. Sterilisation is used to kill almost all micro cellular activity on an item and is rated on the Sterility Assurance Level (SAL) which indicates the probability of a single microbe surviving. Disinfection is used to kill enough micro cellular life to achieve a safe enough level required by the laboratory. Examples of sterilisation and disinfection techniques are as follows:

- Sodium hypochlorite(bleach), is useful as a broad spectrum disinfectant and is effective against bacterial spores[13], [14], [15].
- hydrogen peroxide, which is a very effective sterilising chemical[13] and is often used for sterilisation of tool, biological safety cabinets and laboratories[16].
- Quaternary Ammonium Compounds, which provides cost effective but low level disinfection[13], [14], [15].
- Alcohols, which offer an intermediary disinfection [13], however are ineffective against spores[14]
- Autoclave, which provides extremely effective sterilization using a pressurized containment vessel filled with steam heated to approximately 120°C

The aforementioned commercial solutions specify the use of concentrated alcohol solutions such as ethanol or isopropanol, which are not suitable for adequate sterilisation of microbes which produce spores, such as Mycobacterium tuberculosis, which is the primary cell considered in this feasibility study. Furthermore, Most of these solutions do not specify whether they are Autoclave compatible, leaving compatibility with autoclave sterilisation uncertain. However, the Tokai hit and Zeis/Pecon incubator specifically mention that they are not compatible with autoclaves.

2.5.3 Key Takeaways

The discussed commercial solutions are incompatible with experiments that require stage top incubation for stand alone, long term, live cell imaging within a BSL-3 lab. This is because of the lack of adequate containment capabilities as well as the incompatibility with disinfection and sterilisation methods required by the microbes of interest.

Chapter 3

Specification Analysis

3.1 Commercial Assessment

The commercial solutions from Zeis/Pecon, Ibidi, Tokai-Hit and Okolab are assessed for their primary method of operation as well as the environmental control specifications provided by each product.

3.1.1 Zeis/Pecon

The incubation system provided by Zeis/Pecon utilises four modules for the control of the four major environmental conditions. A temperature unit for thermal control, a CO2 unit for CO2 concentration control, an O2 unit for O2 concentration control and a humidifying bottle for humidity control.

The heating unit monitors and controls the temperature of the culture sample through heating of the incubation chamber, gas airflow and humidifying bottle.

The CO2 unit monitors and controls the concentration of CO2 being pumped into the incubation chamber by controlling the release of pure CO2 gas into the internal mixing chamber.

The O2 unit monitors and controls the concentration of O2 being pumped into the incubation chamber by controlling the release of Pure N2 gas into the circulating gas in order to displace the O2 present. Thus can only achieve hypoxia conditions.

The humidifying unit operates by bubbling the gas mixture through heated water to achieve a close to 100% RH ro prevent the drying of the culture media.

The Zeis/Pecon incubation system as a whole has the specifications shown in Table 3.1.

Temperature Control Range	Ambient - 60°C (Recommended max 45°C)
Control Accuracy	±0.15%
Control Resolution	Internal Loop: 0.01°C User Parameter: 0.1°C
CO2 Control Range	0.0 - 8.0 vol% (8.1-10 vol% with Uncepecified Reduced Accuracy)
Control Accuracy	±2 vol%
Control Resolution	Internal Loop: 0.01 vol% User Parameter: 0.1 vol%
O2 Control Range	0.0 - 21.0 vol%
Control Accuracy	±0.05 vol%
Control Resolution	Internal Loop: 0.01 vol% User Parameter: 0.1 vol%
Operation	PC or Suitable Microscope required
Compatibility	Petri-Dishes, Chambers, Chamber Plates, Chamber Slides, Multi-Well Plates

Table 3.1: Specifications for Zeis/Pecon Incubation System [1], [2], [3], [4]

3.1.2 ibidi

The incubation system provided by Ibidi utilises three modules to maintain the environmental conditions within in incubation chamber. A temperature unit for thermal control, a CO2/O2 unit for CO2 and O2 concentration control, and a humidifying column for humidity control.

The heating unit monitors and controls the temperature of the culture sample through heating of the incubation chamber.

The CO2/O2 unit monitors and controls the concentration of both CO2 and O2 being pumped into the incubation chamber by controlling the release of pure CO2 and N2 into the gas mixture. The release of N2 displaces oxygen present in the air and so the system is only capable of producing hypoxia conditions. The CO2/O2 unit is also responsible for the heating of the gas mixture as well as the heating of the humidifying column and gas tube.

The humidification control system works by controlling the mix of dry gas with humidified gas which is obtained through bubbling the dry gas flow through heated distilled water.

The Ibidi incubation system as a whole has the specifications shown in Table 3.2.

Temperature Control Range	Ambient + 5°C - 50°C
Control Accuracy	±0.2°C with ±0.5°C uniformity
Control Resolution	User Parameter: 0.1°C
CO2 Control Range	0.0 - 15.0 vol%
Control Accuracy	±0.2 vol% + 3% of Reading
Control Resolution	User Parameter: 0.1 vol%
O2 Control Range	0.5 - 21.0 vol%
Control Accuracy	±0.2 vol% (typical), ±0.7 vol% (max)
Control Resolution	User Parameter: 0.1 vol%
Humidity Control Range	Supply RH - 99% RH
Sensor Accuracy	±2.5%
Control Resolution	0.1%
Operation	On Device Controls or Connected PC
Compatibility	Petri-Dishes, Chamber Slides, Multi-Well Plates

Table 3.2: Specifications for Ibidi Incubation System [6], [7]

3.1.3 Tokai Hit

The incubation system provided by Tokai hit utilises one or two modules to maintain the environmental conditions within the incubation chamber, depending on the parameters that are desired to be controlled. The STXG/STXF module is capable of regulating both temperature conditions and CO2 gas concentration. An additional unit, STX-CO2O2, can be connected to the system to override the gas mixture control, and provide control for CO2 and O2 concentration.

The STXG/STXF unit monitors and controls the temperature of the culture sample through heating of the incubation chamber. Additionally this unit is capable of regulating the concentration of CO2 being pumped into the incubation chamber. The STXG model can regulate the CO2 concentration by controlling the release of pure CO2 into the gas mixture, whereas the STXF module requires a pre-mixed pressurised air supply with 5% CO2.

O2 concentration can be achieved using a separate optional module which provides full gas concentration control.

Humidity in the incubation chamber is controlled through a heated water bath that is situated inside the chamber, and so forces a constant high relative humidity.

The specifications for the STXG and STX-CO2O2 incubation system is shown in Table 3.3.

Temperature Control Range	Ambient +5°C - 65°C
Control Accuracy	±0.3°C at heater surface
Control Resolution	User Parameter 0.1°C
CO2 Control Range	5% vol - 20% vol
Control Accuracy	±0.1% vol
Control Resolution	User Parameter 0.1% vol
O2 Control Range	0.1% vol - 18% vol
Control Accuracy	—
Control Resolution	—
Operation	On Device or Connected PC
Compatibility	Petri-Dishes, Chamber Slides, Multi-Well Plates

Table 3.3: Specifications for the Tokai Hit Incubation System [8]

3.1.4 OkoLab

The incubation system provided by OkoLab utilises three modules to maintain the environmental conditions. A temperature unit for thermal control, a gas unit for CO2 and O2 concentration control and a humidifying bottle.

The heating unit monitors and controls the temperature of the temperature of the culture sample through heating of the incubation chamber.

The CO2/O2 unit monitors and controls the concentration of both CO2 and O2 being pumped into the incubation chamber. The gas module is capable of working with a mix of concentrated O2, concentrated CO2, and concentrated N2 to obtain a complete control of the gas mixture being passed into the incubation chamber. Alternatively, the unit can operate with a supply of pressurised air, concentrated N2 and concentrated CO2 to get less complete atmospheric control.

Humidity control is obtained by bubbling the gas mixture through heated water to increase the relative humidity.

The specifications for the OkoLab incubation system is shown in Table 3.4

Temperature Control Range	Ambient +3°C - 60°C
Control Accuracy	±0.1 for sample feedback, ±0.3 for chamber feed-back
Control Resolution	User Parameter 0.1
CO2 Control Range	0% vol - 20% vol
Control Accuracy	±0.2% vol at 5% vol
Control Resolution	User Parameter 0.1% vol
O2 Control Range	0% vol - 95% vol (using Pure O2 source, otherwise 20% max)
Control Accuracy	±0.2% at 5% vol
Control Resolution	User Parameter 0.1% vol
Humidity Control Range	26% RH - 95% RH at 50
Control Accuracy	±1%
Control Resolution	User Parameter 1%

Table 3.4: Specifications for the OkoLab Incubation System

3.2 Requirements

The final goal of this project is the development of a stage top incubator that can be used to do long term observations of M. tb without the requirement of a large external incubation chamber.

However this report focuses on determining the feasibility of undertaking this project, and so has a reduced set of requirements aimed at scoping the fundamental aspects that would be required to understand the project goal. Therefore the requirements for this project include the following:

- Containment of Mycobacterium tuberculosis (M. tb)
- Compatibility with low magnification microscope stage
- Compatibility with a BSL-3 laboratory
- Compatibility with a 96-well culture vessel

This project focuses on the main aspect of bio-containment within a BSL-3 lab and how that can be achieved with a format that is compatible with the growth vessels and microscopy standards that are utilised for this kind of experiment.

3.3 Specifications

The specifications for the commercial solutions were assessed to obtain an understanding of specifications this project would need to meet in order to achieve a incubation system that would provide suitable conditions required for this type of long term imaging with cell cultures.

Table 3.5 shows the list of specifications that were decided on based off the specifications for the commercial solutions shown above. These specifications fall within those provided by commercial

Temperature Control Range	Ambient - 55
Temperature Accuracy	0.2
Temperature Resolution	0.1
CO2 Control Range	0% vol - 15% vol
CO2 Control Accuracy	0.1% vol
CO2 Control Resolution	0.1% vol
O2 Control Range	0.5% vol - 20% vol
O2 Control Accuracy	0.1% vol
O2 Control Resolution	0.1% vol
Humidity Control Range	90% RH - 99%RH
Humidity Control Accuracy	—
Humidity Control Range	—

Table 3.5: Derived Specifications for the Continuation of this Project.

solutions and should facilitate the incubation of most cell cultures.

3.4 Feasibility

discus whether undertaking this project is feasible.

Chapter 4

Conceptual Design

The main focus on this report is centred about achieving a bio-containment system that allows for observing cultures grown in a 96-well plate. The system must ensure that no contamination can occur from the cells grown inside the container.

This section aims to describes the steps and decisions taken in the process of making a conceptual design for a containment vessel that meets the requirements mentioned in chapter 3. This conceptual design will help to determine the feasibility of taking this project further and incorporating incubation techniques which will enable the bio-containment vessel to be utilised as an incubation chamber that can be used for long term experimentation independently from traditional laboratory incubation cabinets

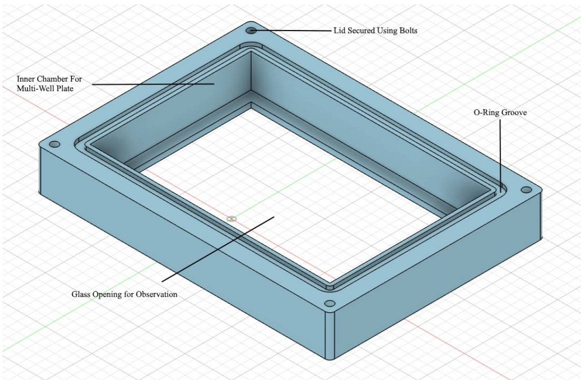
4.1 Design Iterations

4.1.1 Design Version 1

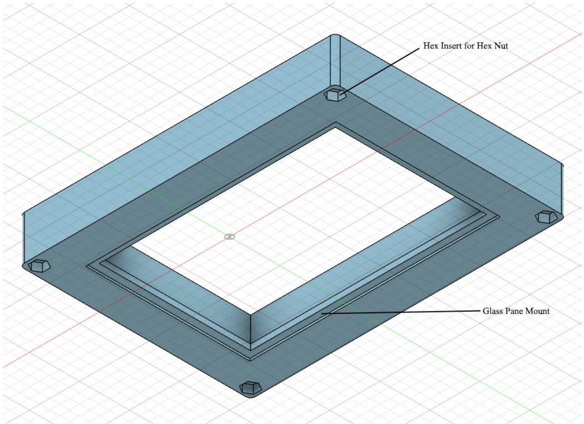
The first design version uses a base and a lid to seal the multi-well plate inside the chamber.

As can be seen in Figure 4.1a, the base of the chamber has a internal cavity, which allows a multi-well plate to be placed inside, the plate is designed to rest on a ledge along the bottom of the inner cavity. There is a groove along the top which is designed to align with a groove in the lid shown in Figure 4.2b, and house an O-Ring which will seal the inner chamber. The base is designed to mount with the lid with the use of four M4 bolts which will run through the four holes in the corners of the base and lid.

Figure 4.1b shows the recess which is designed to allow a glass plate to be embedded into the base and sealed using epoxy or glue. This allows for a clear optical path for viewing the growth culture. The

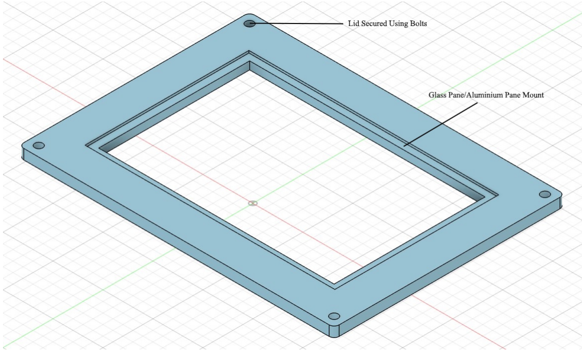


(a) Conceptual Design 1 Base Top-Side

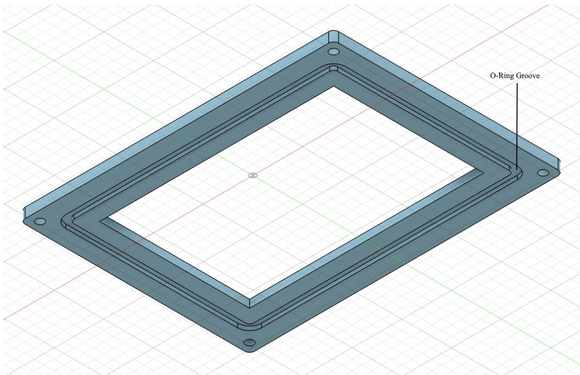


(b) Conceptual Design 1 Base Bottom-Side

Figure 4.1: Conceptual Design 1 Base



(a) Conceptual Design 1 Lid Top-Side



(b) Conceptual Design 1 Lid Bottom-Side

Figure 4.2: Conceptual Design 1 Lid

base also includes hex shaped recesses which allows for embedding a hex nut to interface with the M4 bolts used to mount the lid.

As shown in Figure 4.2a, there is a similar recess which is designed to allow a glass plate or aluminium plate to be embedded into the lid sealed using epoxy or glue. A glass plate can be used to allow for a complete optical path from the top to bottom of the chamber, alternatively an aluminium plate can be embedded into the lid, this plate would utilise LED's and a heating element to provide better heating capabilities, as well as allowing the chamber itself to provide the required illumination required for microscopy.

Changes

This design was improved upon using the following alterations.

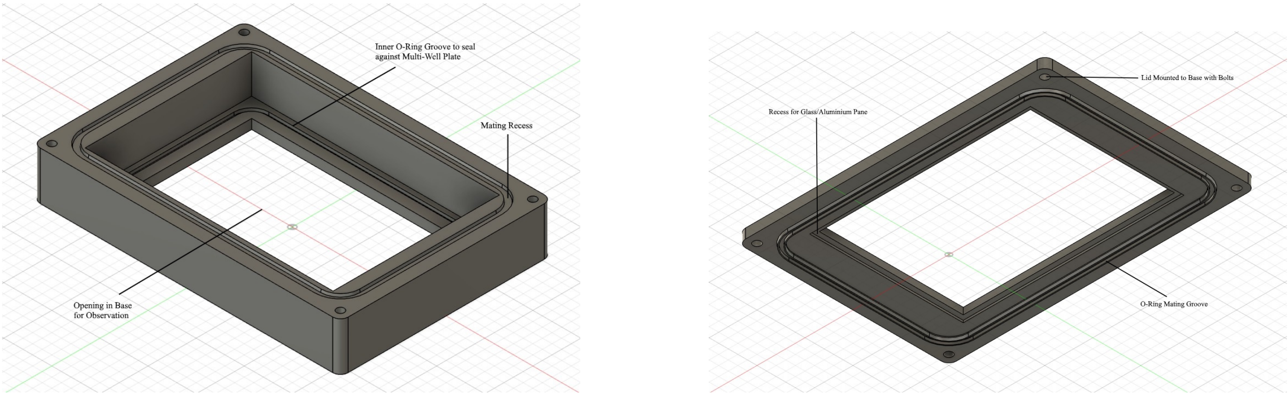
The recess for the glass/aluminium pane on the lid was moved from the outside of the lid to the inside of the lid. By making this change, any potential pressure within the container will press the pane into the lid and ensure the seal.

The design for a glass pane on the base of the container was removed and the frame of the multi-well plate was used as the sealing agent for the base of the container. By removing this pane on the base, there is a reduction in the chance of condensation forming on the pane which may obscure observation. Additionally, this allows the floor of the base to be thinner, which allows the multi-well plate to sit closer to the microscope objective and will allow for better observation.

The O-ring groove was altered to use a mating feature that allows the lid to mate with the base to ensure that there is proper alignment of the two parts. Additionally the O-ring would rest in this feature and utilise a standard O-ring groove design.

4.1.2 Design Version 2

The fundamental design for the second version of the conceptual design is similar to that in version one. The chamber uses a base with a detachable lid to seal the multi-well plate inside the container. Two o-rings are used to seal the chamber and prevent contamination.



(a) Conceptual Design 2 Base (b) Conceptual Design 2 Lid

Figure 4.3: Conceptual Design 2

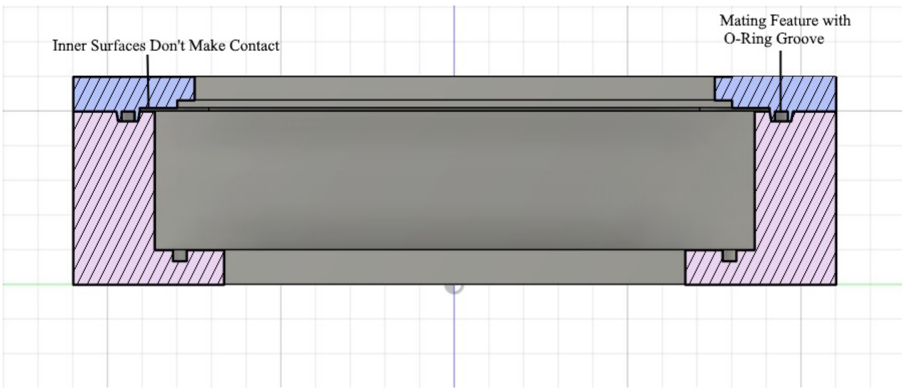


Figure 4.4: Cross Section of Conceptual Design 2 Assembly

O-Ring

As can be seen in Figure 4.3 and Figure 4.4, this updated version uses a mating feature where the base incorporates a recessed groove which will interface with the extruded feature on the lid. This ensures that the lid and base align correctly to provide a better seal.

The groove for the o-ring was designed into the mating feature to ensure that the o-ring makes consistent contact with the intended sealing surface, which will be within the recess present on the base.

The base and lid feature a parametric design model, where the parameters of the o-ring to be used can be entered and the mating feature and o-ring groove will dynamically change to suit the given parameters

The total perimeter of the feature is determined by the circumference of the o-ring. The circumference of the o-ring is approximated by using Equation 4.1

$$Circumfrence = 2\pi(\frac{Oring_{ID}}{2} + \frac{Oring_{OD}}{2}) \tag{4.1}$$

Based on this circumference, the O-ring groove is changed such that the it maintains the same rectangular ratio as the chamber itself, whilst changing the total mid-point perimeter such that the o-ring is not stretched when placed inside the groove.

The groove itself is designed according to recommendations and guides provided by the Parker Handbook [17] and online articles including O-Ring Groove (Gland) Design Guide by Tessa Axsom [18]. The width of the o-ring groove is designed to allow for a 5% expansion in the o-ring diameter [18], this will ensure that the o-ring gets sufficiently compressed by the walls to ensure a suitable seal. According to the guides provided by Marco sealing solutions [19] and Tessa Axosm [18], in order to obtain a suitable sealing effect, a static face seal must use an o-ring with a compression ration of between 20% and 30%.

$$CompressionRatio = \frac{ORing_{Diameter} - GrooveDepth}{ORing_{Diameter}} \tag{4.2}$$

$$GrooveDepth = ORing_{Diameter}(1 - CompressionRatio) \tag{4.3}$$

The equation for the compression ratio of an o-ring shown in Equation 4.2

The groove depth is designed to be 20% smaller than the diameter of the o-ring [17], [18], [19]. This allows for the o-ring to be sufficiently compressed to provide a proper seal.

4.2 Final Conceptual Design

The final design is the following, and uses the following features.

Chapter 5

Discussion and Results

Chapter 6

Conclusion

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6.1 Recommendations

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